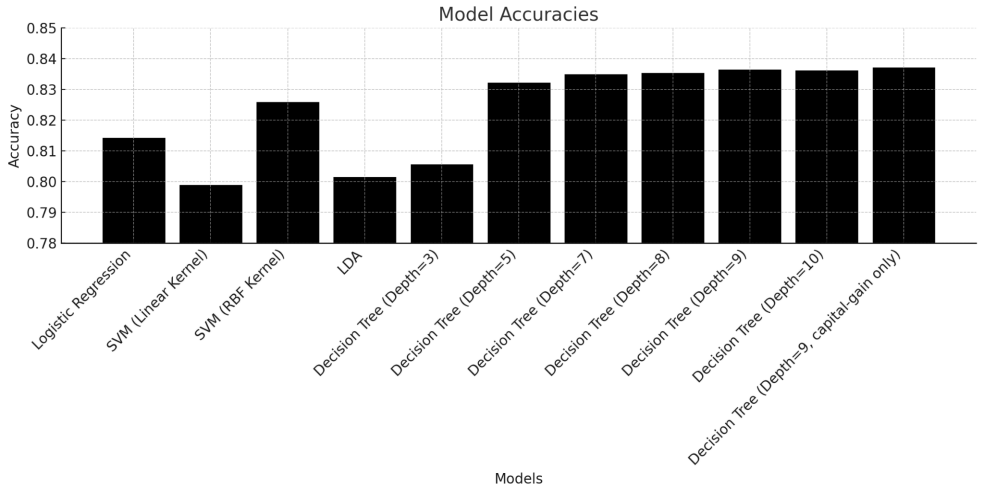
DTSC 2302 Project 1 Report

Aidan Hall, Ruhan Rahman, Mason Davis, Will Jones

Part A

The data used for our model includes features that we will use to predict whether an individual's salary is less than or equal to 50k or greater than 50k. The features we used to predict this include age in years, years of education, amount of capital gains in the last year, amount of capital loss in the last year, and hours worked per week. Based upon cross-validation, we will be testing the following models: logistic, linear discriminant analysis, decision trees, and support vector classifiers.

We tested all of these models (tested both linear and RBF kernels for SVM and depths 3, 5, and 7 for decision tree) with the entirety of the features. What we found was that the decision tree had the best accuracy, specifically with a depth of 7. We continued to raise the depth until the accuracy decreased. The accuracy continued to increase until depth 10, which left us with a depth 9 decision tree as the best model. Lastly, we looped through all combinations of features and found that only using the amount of capital gains in the last year as a predictor yielded the greatest accuracy at 83.69%.



1. Above is Figure 1, a bar chart representing the accuracies of each model we tested. The X-axis has the names of the model underneath each bar with accuracies as a percentage represented on the y-axis. On the left side (everything up to decision tree depth 3), it is very clear that the decision trees with a depth of 5 and above dominate in accuracy. It's a little harder to tell the difference on the right side, but the accuracy increases until depth 10, where we then go back to depth 9 and find the accuracy with the best combination of features, which was only capital gain.
2. We recommend using a decision tree model. You will see the best accuracy with a decision tree depth of 9 and only using capital gain as a feature. This is 83.69% accurate, which avoids overfitting since it has been cross-validated.
3. Using the trained model, we made predictions for two hypothetical individuals. The first individual is 39 years old, has completed 10 years of education, reports no capital gains or losses, and works 40 hours per week. The model predicted that this individual earns less than or equal to $50,000 annually. In contrast, the second individual is 45 years old, has completed 14 years of education, reports $2,000 in capital gains, has no capital losses, and works 35 hours per week. For this individual, the model predicted an income of over $50,000 annually. These predictions suggest that the presence of capital gains may positively influence the likelihood of earning above $50,000, even if the number of hours worked per week is slightly lower.

Part B:

For this part, we decided to make a model from the data from credit cards and the data that Is included in this model is the gender the person is and what types of college education the person will have. The model will also include age and whether you are married or not.

Now looking at the mode we made is a logistic model and we looked at stuff like decision tree and Bootstrapping. We decided to choose a logistic model because first It gives me a breakdown of the entire variable and their chances of paying next month's credit card bill. We also chose it because we can show you how accurate the entire model is so we can give the people their confidence.

```
                           Logit Regression Results
==============================================================================
Dep. Variable:     default payment next month   No. Observations:               30000
Model:                                  Logit   Df Residuals:                   29996
Method:                                   MLE   Df Model:                           3
Date:                        Mon, 07 Apr 2025   Pseudo R-squ.:                0.002718
Time:                                02:03:22   Log-Likelihood:                -15810.
converged:                               True   LL-Null:                       -15853.
Covariance Type:                    nonrobust   LLR p-value:                 1.460e-18
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.9252      0.074    -12.539      0.000      -1.070      -0.781
SEX           -0.2005      0.028     -7.105      0.000      -0.256      -0.145
EDUCATION      0.0765      0.018      4.364      0.000       0.042       0.111
MARRIAGE      -0.1020      0.027     -3.785      0.000      -0.155      -0.049
==============================================================================
Accuracy: 0.7788
```

Looking at our model above, the first aspect we'll examine is its accuracy. The model has an accuracy rate of 77.9%, meaning it correctly predicts whether individuals will make their credit card payments for the next month approximately 78% of the time.

Next, we see the coefficient for the variable Sex, which is -0.20. This tells us that being a particular gender is associated with a 20% decrease in the likelihood of making the next month's payment.

The following line in the logistic regression output shows the standard errors, which are all relatively low. This indicates a high level of confidence in the model's estimates.

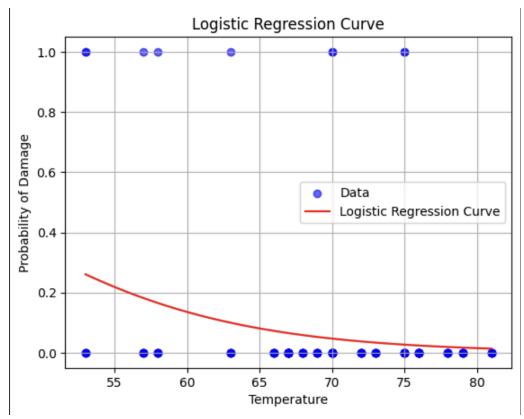
Finally, we can look at the z-scores. These are fairly spread out, which tells us that some variables are stronger predictors of payment behavior than others, contributing to the overall accuracy of the model.

Part C:

For Part C, we created a logistic regression model. Firstly, we converted the "Damaged" column to a numeric state. Taking a look at our coefficients, -5.085 is the predicted log odds of an O-ring being damaged. Furthermore, for each additional degree of temperature, the predicted log odds of an O-ring being damaged decreases by -0.116. Next, to get the predicted probabilities, we created a new X and added a row of constants that only contained the three temperature values we wanted to predict for (50, 40, 30). We got the following results from our model prediction: Temp 30:

1. probability of O-ring failure (is Damaged) = 0.834
2. Temp 40: probability of O-ring failure (is Damaged) = 0.613
3. Temp 50: probability of O-ring failure (is Damaged) = 0.33

```
                           Logit Regression Results
==============================================================================
Dep. Variable:        Damaged_Numeric   No. Observations:                  138
Model:                          Logit   Df Residuals:                      136
Method:                           MLE   Df Model:                            1
Date:                Sun, 06 Apr 2025   Pseudo R-squ.:                 0.09234
Time:                        14:52:19   Log-Likelihood:                -30.198
converged:                       True   LL-Null:                       -33.270
Covariance Type:            nonrobust   LLR p-value:                   0.01319
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          5.0850      3.052      1.666      0.096      -0.898      11.068
Temp          -0.1156      0.047     -2.458      0.014      -0.208      -0.023
==============================================================================
0    0.834373
1    0.613235
2    0.332904
dtype: float64
```



Our prediction results affirm our findings that as temperature increases, the probability of O-ring damage decreases. We can further observe this relationship by taking a look at the graph we generated with the predicted logistic regression curve. From the graph, we can see that the scatter of points skews towards "un-damaged" on the side expressing higher temperatures. We can also see from the red curve line that as the temperature increases, the probability of damage decreases at an exponential rate. Overall, our graph and model allow us to see the negative logistic relationship between Temperature and O-ring damage.

Ethical Analysis

For us, the phrase "data scientists should never surrender their professional judgment to organizational pressure" means that we must uphold our integrity regardless of whether or not it contradicts the organization. When describing a situation like the Challenger disaster, giving in to pressure may lead to errors such as misunderstood calculations, incorrect data usage, and unfactored risks. As a data scientist, it is our responsibility to ensure that we analyze every possible factor and outcome accurately and assess each risk to determine how successful the mission will be.

Consequences of violating this principle can lead to massive failures, such as the Challenger exploding mid-flight. Engineers were concerned with the O-rings failing at low temperatures, but the organization pressured the mission to continue without delay. This ultimately led to the loss of seven lives and highlights the importance of patience and accuracy, as well as ensuring all risks are properly accounted for. As we progress further into the field of data science, it is imperative we hold our ethical practices to a high standard to safeguard the good of the public and protect those affected by our analyses and maintain trust with decisions relating to data.

Asking data scientists to prove a launch is safe rather than proving it is unsafe creates an uncomfortable ethical problem. This perspective favors proceeding with known risks involved, overlooking possible errors to complete the mission. This allows for decision makers to validate an outcome without assessing every potential flaw in a plan, and can lead to concerns being overlooked and a false sense of security being established.

This relates to confirmation bias in a way that people tend to seek validation for their existing beliefs or goals. This includes only looking for evidence that favors chances of success

and ignoring evidence that may point to potential failure. If a data scientist were to only focus on proving something is safe, they may overlook contradictory evidence or be reluctant to disclose information that could raise doubts. Ethical risk communication needs transparency, honesty, and the ability to include uncertainty. With a different approach or perspective, data scientists should consider every angle of a problem and evaluate the likelihood of failure. When faced with organizational pressure, proper communication of risks and taking a more cautious approach can improve the rate of success and might even save lives.

The reliability of technology before being used in high-risk missions must be very high. Human lives, expensive assets, and lasting effects are at great risk if technology is not reliable and should be tested, validated, and reviewed before anything takes place. Simulations shall be run to ensure minimal failures and should include contingency plans for safety and protection. In addition to testing, continuous monitoring and data analysis should be implemented to identify and address potential issues.

Ethical considerations that guide this threshold include human safety, informed consent, transparency, and accountability. It is ethically responsible to minimize the risks to human life and environmental harm, ensuring that the probability of failure is extremely low. Informed consent refers to disclosing risks, uncertainties, and safety measures to the individuals directly impacted by the use of technology. This also includes evaluating how comfortable everyone is with the specific technologies implemented. Some of the most important features of ethics within data science are transparency and accuracy. Being transparent with the selection and usage of certain technologies mitigates risks and enhances comfortability. Individuals must be held accountable for any possible errors or failures so that actions can be taken to resolve any issues.

The threshold for using technology in high-risk missions such as spaceflight should be based on high reliability, extensive testing, and ethical considerations regarding human safety and accountability. With greater risk involved, technology must be more reliable to ensure the best chance of success under any circumstances.

Conclusion

To conclude, our analysis highlights the value of data science models to assess income, credit card payments, and the safety of space missions. By testing models such as decision trees and logistic regression, we identified key predictors to optimize accuracy. This also emphasizes the importance of ethical considerations within data science. We've discussed the risks of organizational pressure, the difference between proving safety versus proving risk, and the need for transparency and accountability whilst using reliable technology. For high-risk missions such as the Challenger mission, it is mandatory for technology to undergo heavy testing and for risks to be calculated to mitigate potential errors. Ultimately, including all of these considerations into our data science practices not only increases model accuracy but ensures safety, trust, and mission success.